

Lathe Theory

Intro to the Lathe Machine

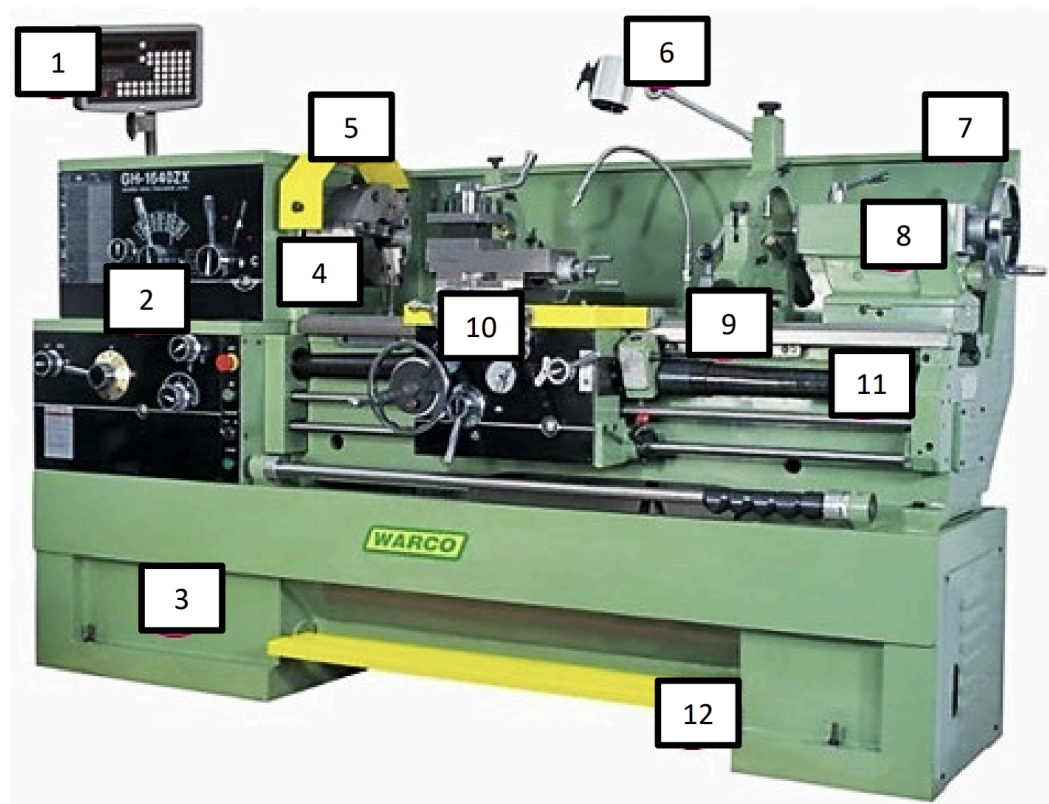
A lathe is a machine that does a turning process; the process of removing material from a workpiece, or work, with a single pointed tool.

A single pointed cutting tool is a tool with one dominant edge that removes material.

For a lathe, the workpiece is turned while the tool is moved cross wise and lengthwise to the axis of the bed, i.e. along the bed/along X axis (left/right) and Z axis (towards work)

Parts of a Lathe

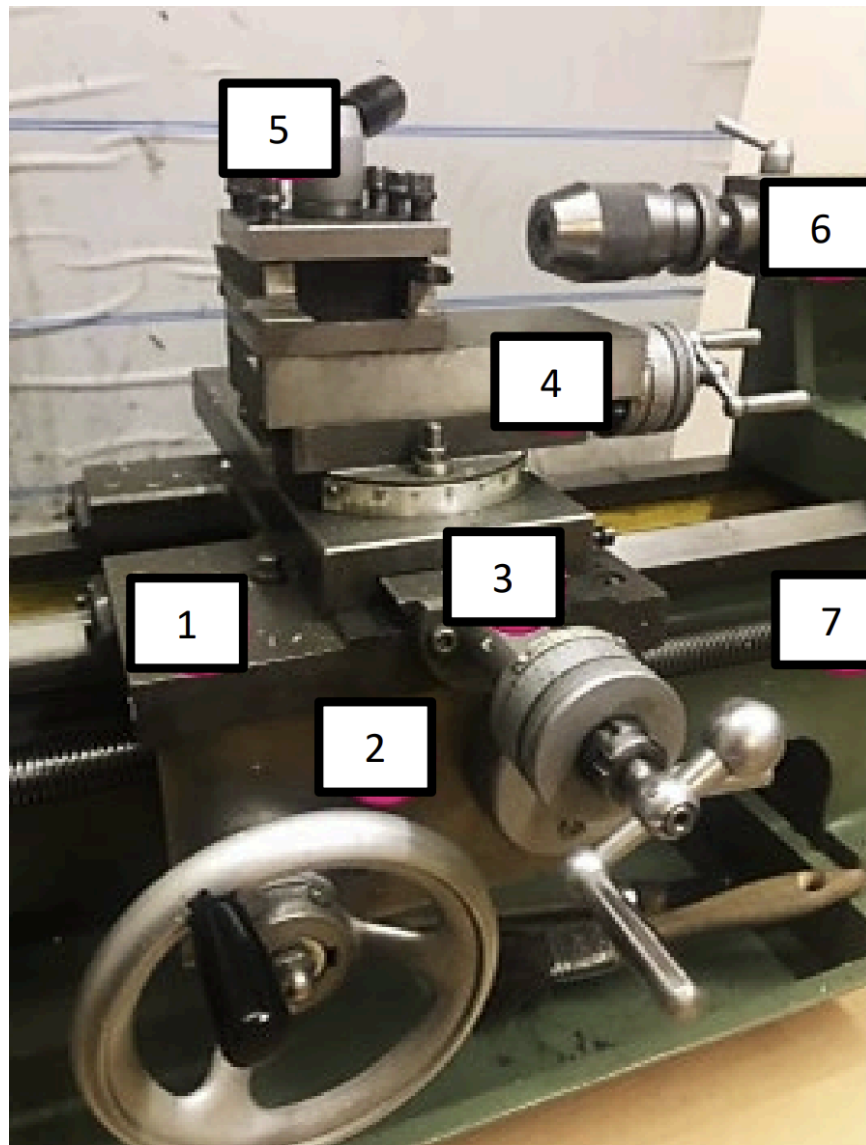
Shown is a conventional lathe machine:



	Part Name	Purpose
1	Digital Readout (DRO)	Indicates location of tool along X and Z axis, allowing it to reset position and for accurate cutting.

2	Headstock	fixed part that holds motor, gearbox, spindle and controls for the spindle and lead screw rotational speed.
3	Support	supports and raises lathe
4	3 Jaw chuck	securely holds and rotates the workpiece at high speeds, due to being connected to the spindle. the spindle can rotate in either direction
5	Safety Guard	prevents contact from rotating chuck. when flipped open, the chuck key sockets can be accessed, and prevents the spindle from turning on
6	Light	provides illumination
7	Splash Guard	prevents coolant and chips from spilling onto the floor
8	Tailstock	movable along bed while maintaining center position to spindle center. holds tools. in the tailstock is a barrel, that can be extended or retracted via a handwheel. the tailstock and barrel can be locked in place via levers at the end of the tailstock
9	Bed	base of lathe machine; mounted to headstock and allows tailstock to slide on it. length of bed determines size of workpiece machined. bed also contains lead screw running through it.
10	Carriage	moves between headstock and tailstock; holds components to move the tool, the handwheel to move carriage along bed, and control lever to start the lathe. carriage movement controls depth of cut and can also be engaged to lead screw.
11	Lead Screw	connected to motor spindle, allows carriage to be engaged to it anywhere along its length, to move it at a specific rate (e.g. for threads)
12	Brake Pedal	switched off/brakes spindle; acts as emergency stop.

The carriage is the main thing that is controlled during turning operations, and has more intricate parts:



1	Saddle	top of carriage; slides on tool bed and supports cross slide.
2	Apron	front of carriage; houses mechanism to engage carriage with lead screw, holds lever to start and stop machine and hand wheel to move carriage along the bed.
3	Cross Slide	mounted on saddle, allows compound slide to move perpendicular to spindle via its hand dial, to control the depth of cut during a turning operation

4	Compound Slide	mounted on cross slide; allows tool post to be swivelled and moved via its hand dial, for tapered features, depth of cut for facing, or for minor turning operations.
5	(4 Way) Toolpost	hold tools. 4 way refers to it being able to hold 4 tools.
6	Tailstock (with Drill Chuck)	same as tailstock from earlier, however the drill chuck helps to hold drill bits, and can be tightened or loosened with its drill chuck key.
7	Lead Screw	

Types of Motion

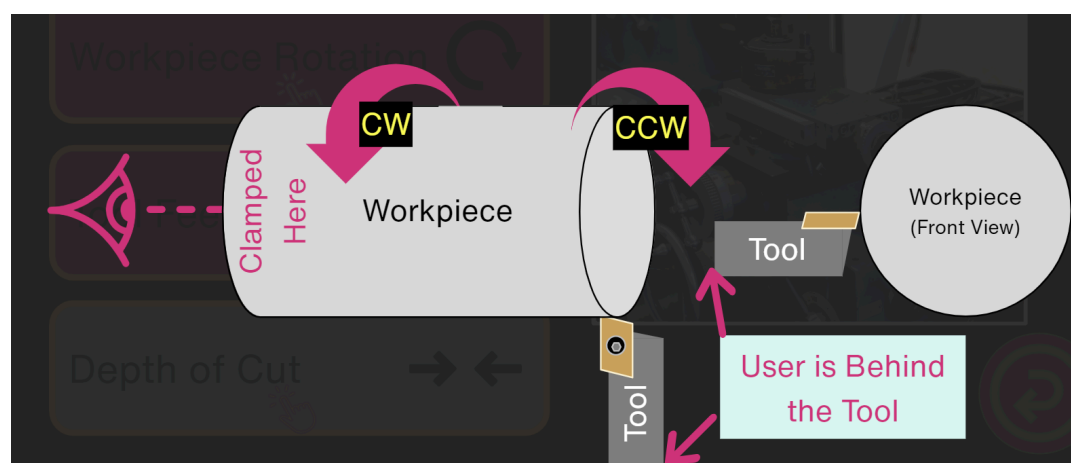
The lathe can undergo different types of motion, for different types of cuts and operations.

Workpiece Rotation

The workpiece can be rotated clockwise or counter clockwise; its direction determined by looking from the chuck to the workpiece (clockwise: rolls towards, anticlockwise: rolls away)

The direction is important based on the type of tool used, whether it is left or right handed.

The workpiece must rotate against the cutting edge of the tool, otherwise the work will rub against the tool instead.



The diagram depicted is a right handed tool (RHT), because its easier to control with the right hand, not that its cutting edge is on the right.

Notice how for RHTs, the workpiece has to rotate clockwise. Thus, left handed tools (LHTs) must have the work rotate anti clockwise.

Tool Feed Direction

The tool can be moved either longitudinally (left/right along length of work) or radially (inwards/outwards along radius of work)

Longitudinal Feed Directions (Z axis) are useful for reducing the diameter of the workpiece, by moving towards the chuck to remove material.

Radial Feed Directions (X axis) are useful for reducing the length of the workpiece, by moving it from outside the workpiece towards the center to remove material.

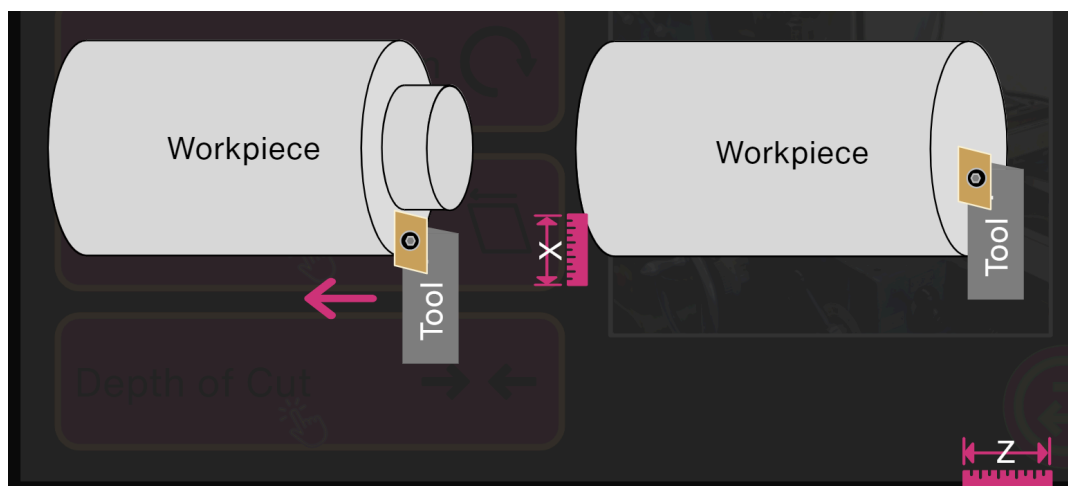
Depth of Cut

For Radial Cuts/Cuts that reduce the length of the workpiece, the tool is moved along the z axis first to dictate how much of the workpiece will be machined off. (this dictates the depth of cut)

Afterwards, the tool then moves radially inwards to engage and cut the workpiece.

The depth of cut cannot be too high, as the tool may be unable to cut a large portion of the work all at once.

To reduce the diameter of the workpiece, the tool is moved along the x axis first to dictate the depth of cut, then moved longitudinally to cut the workpiece.

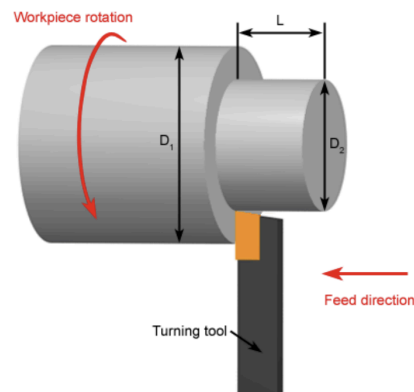


With different types of motion, the lathe can be used for different operations on either the exterior (external operations) or the interior (internal

operations) of the work.

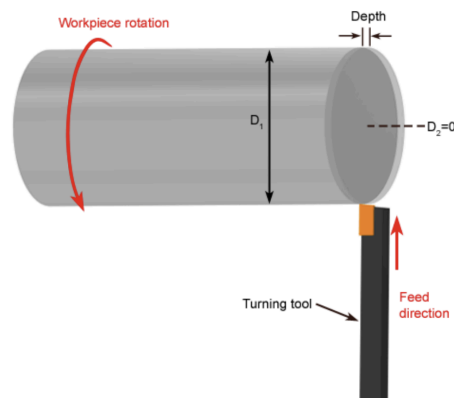
External Operations

Turning



turning tool is fed in the z direction to reduce the external diameter of the work

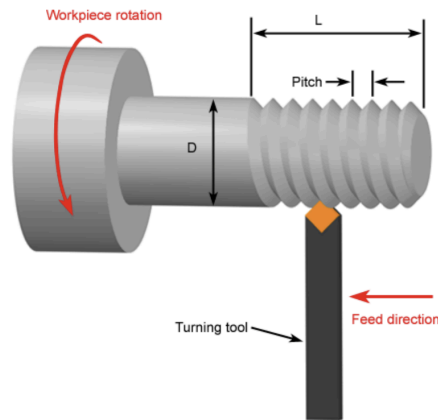
Facing



turning tool (usually same tool is used for turning and facing) is fed in x direction to reduce the workpiece length

Other specialised (and non interchangeable) turning and facing tools may be used instead.

Threading

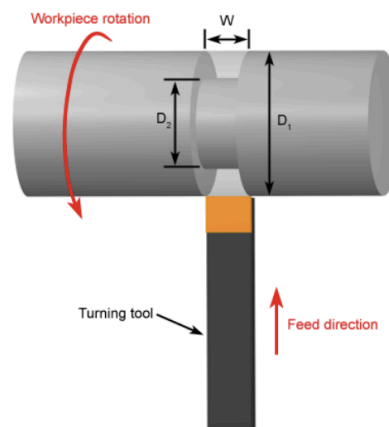


A threading tool/thread die is fed along the z axis to create external threads.

The workpiece is rotated at low speeds with high amounts of lubrication, while the tool is fed at a faster rate for the spiral threads around the whole workpiece (and constant rate due to carriage apron being engaged with the lead screw)

As the tool is fragile, the workpiece must be turned to correct diameter and surface finish before threading.

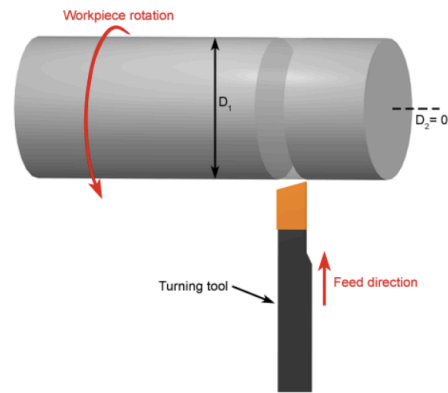
Grooving



A grooving tool is fed in the x direction to create grooves in the workpiece, and is fed at a slower rate.

These grooves usually do not go all the way into the center of the workpiece, and are used instead for easier assembly, or to connect things like circlips.

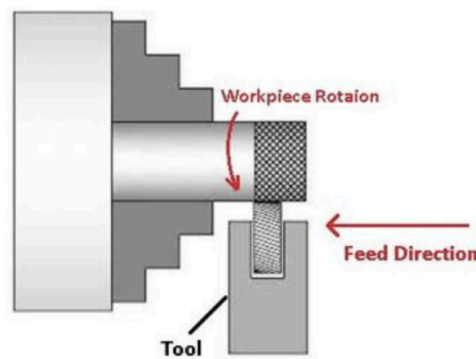
Parting



A parting tool is fed in the x direction similarly to grooving, however it plunges to the center, to part/cut off a section of the workpiece.

Again, the parting tool is fed at a slower feedrate.

Knurling



A knurling die is fed in the z direction to impress/form patterns(diamond, straight, angled) on the metal surface, for a better gripping surface

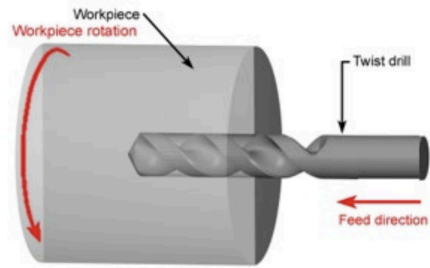
As knurling is also a forming process (metal is deformed, not cut), it is done at low speeds.

Other operations like chamfering, radius forming, knife turning, may also be used.

Internal Operations

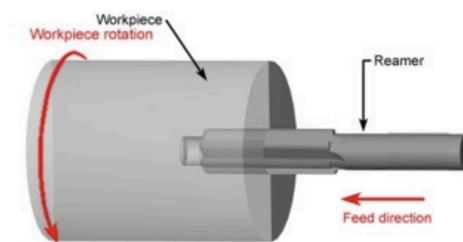
These operations usually have the tool mounted on the tailstock, instead of the 4-way toolpost.

Drilling



The drill bit is fed along the z direction into the work piece, to create a hole.

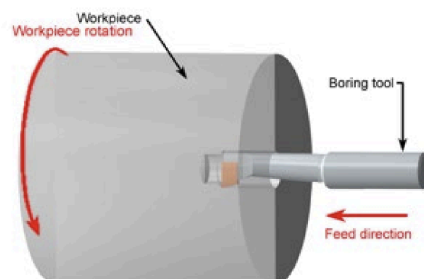
Reaming



The reaming tool is fed along the z direction into the work, to create a highly accurate hole with a smooth surface finish (from a pre-existing hole via Drilling).

This operation is done at slow speeds with large amounts of lubrication

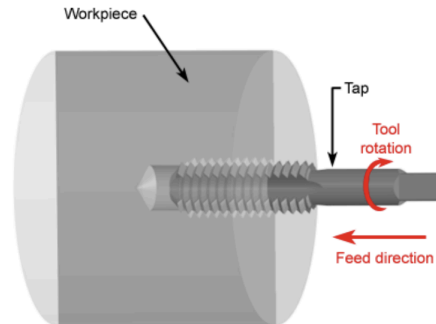
Boring



The boring tool is fed in the z direction into the work, to enlarge an existing hole, or to finish an existing hole to high accuracy (with smooth surface finish)

The existing hole must also be bigger than the boring tool, so that it can fit inside to enlarge the hole

Tapping



A thread tap is fed in the z direction into the work to form internal screw threads from an existing hole. The tap size used is dependent on the size of the hole.

Like threading, this operation requires low speeds and high lubrication.

Metal Cutting Theory

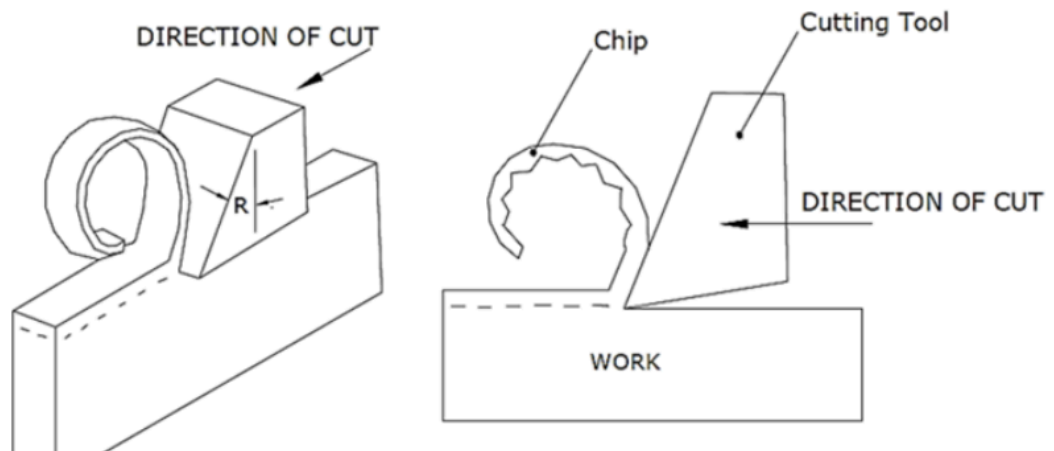
Cutting Methods and Forces

Orthogonal Cutting

Orthogonal cutting refers to when the cutting edge is perpendicular to the direction of tool movement.

This causes chips to form into flat spirals on top of the cutting edge, which may not flow away easily, making chip disposal harder.

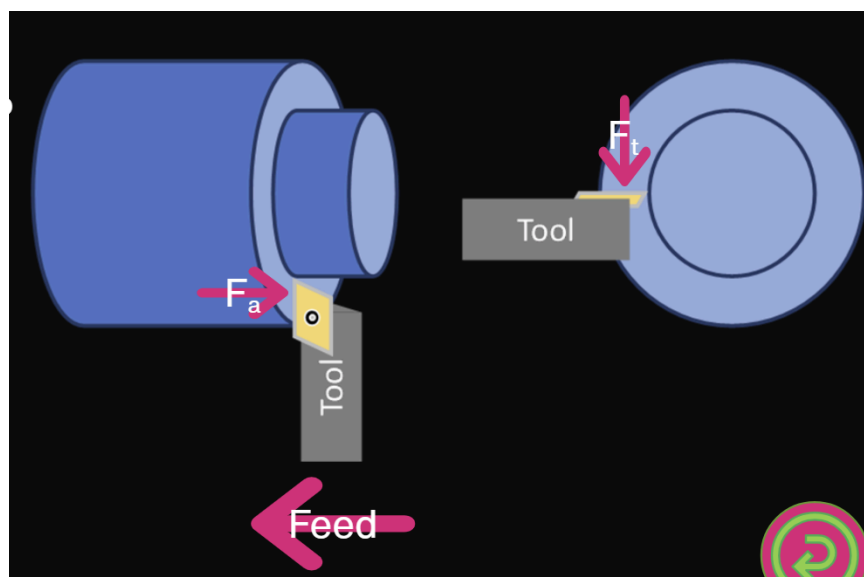
This cutting method may also cause the workpiece surface to be damaged.



feed direction is along z axis, cutting edge is along x axis. However, orthogonal cutting is still necessary for operations like shoulder turning (similar to turning, however for making a sudden change in diameter) and grooving.

As the chips coil on top of the tool, they may get stuck (e.g. in the groove during grooving), thus orthogonal cutting must be done slowly. (due to poor chip removal, chip clogging, and higher cutting force needed).

During Orthogonal Cutting, when the tool is fed into the workpiece, it experiences two forces, the Axial and Tangential Force.



The axial force, F_a , acts along the turning axis, opposing the feed direction.

this is similar to a normal reaction force, and is much smaller and not as significant than the tangential force.

The tangential force, F_t , acts tangential to the cutting surface, opposing the cutting action.

This force is much stronger and significant, causing it to consume the most power, and may damage/break the tool.

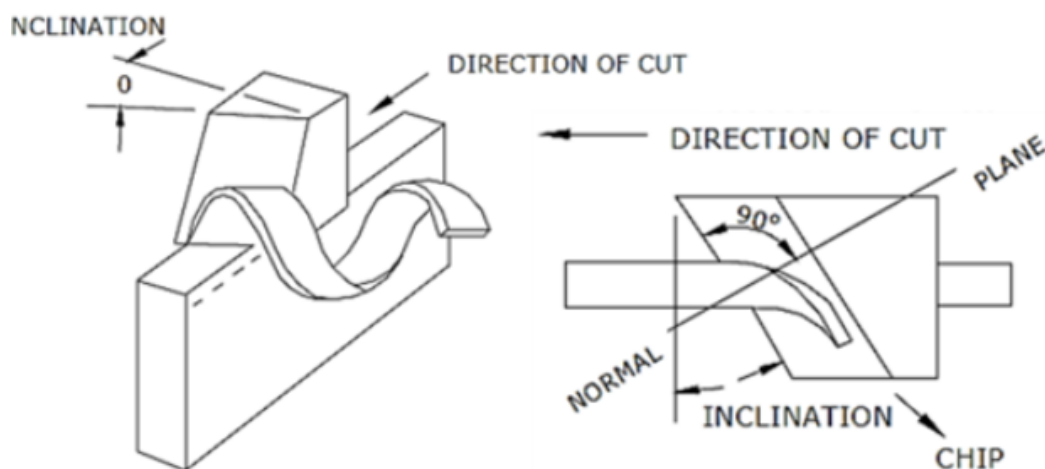
Thus, a lower feedrate/lesser depth of cut should be used.

Oblique Cutting

Oblique cutting has the cutting edge at any angle that is not perpendicular to the feed direction.

This causes chips to form into a spiral coil, allowing it to flow away without disturbing the cutting surface, giving it a better surface finish

As lower force is needed for oblique cutting, deeper cuts are possible with this method.

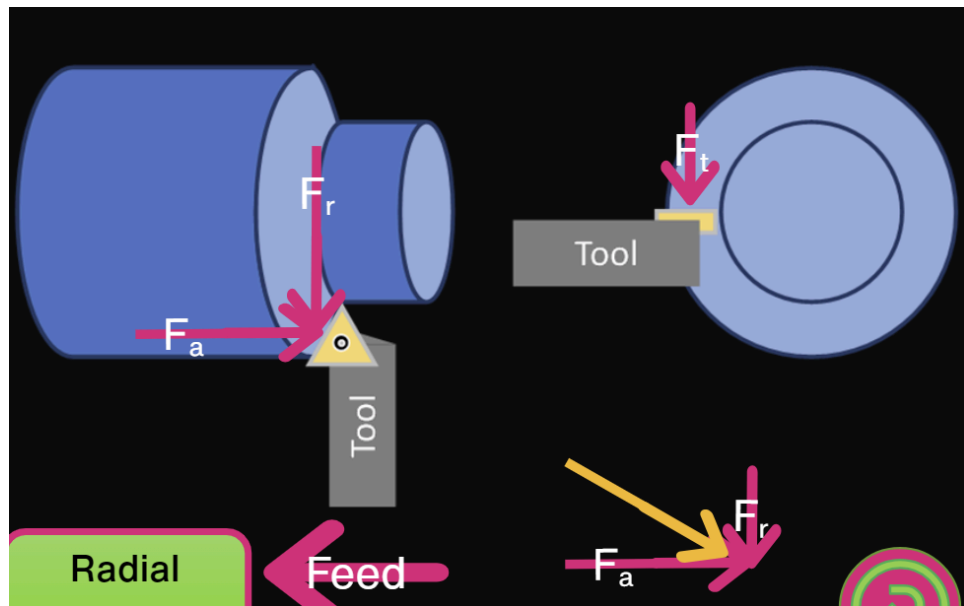


Cutting edge is inclined from the x axis, so that it makes a non-90 degree angle with the z axis.

Because of its benefits, most operations use oblique cutting, like turning or facing, where the chips can spiral into coils and flow away easily.

Due to the lower force needed and the easier chip removal, these operations can be done at faster speeds.

During Oblique Cutting, the tool experiences 3 forces when fed into the workpiece, Axial, Tangential and Radial Forces,

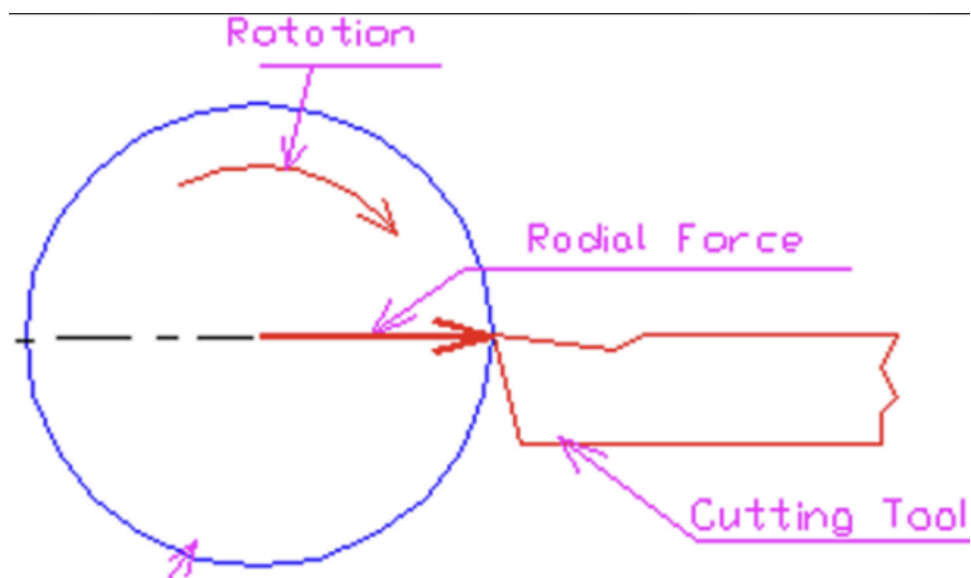


With the cutting edge at an angle, it causes the normal force of the workpiece pushing back on the tool to be split into two forces, the axial and radial forces.

The tangential and axial forces are still the same from orthogonal cutting in oblique cutting, however the radial force, F_r , acts radially, opposing the cross feed direction.

As the radial force is a component of the normal force, it also is a small force, and is not very significant.

Even though the force does push the tool away from the workpiece, the cross slide counters this force and resists the radial force.



Cutting Factors

There are 7 major factors affecting cutting.

Workpiece Material

Depending on the part needed, different materials with different properties (composition of material, manipulated mechanical properties like heat treatment) may be used, thus requiring different tools, cutting conditions and machine power.

Cutting Condition

There are three types of cutting conditions

Cutting Speed

Cutting speed is the workplace surface speed during the cutting operation. in other words, how much of the workpiece's arc length advances in 1 second.

Notice how this is different from rotational speed, as two workpieces (one with 1cm diameter, other with 2cm diameter) can rotate at the same speed, but the 2cm diameter will have the surface advance more.

The range of cutting speed possible (called the working range) is determined by the type of tools used, as well as the workpiece diameter and material.

For rougher/less precise cuts, a lower cutting speed is used
For finer/more precise/surface finishing cuts, a higher cutting speed is used.

Feedrate

Feedrate is the speed of the tool moving across the workpiece surface, and its working range is determined by the type of tool used and the material type

For rougher cuts, a faster feedrate should be used.

For finer/finishing cuts, a slower feedrate should be used.

Depth of Cut

The depth of cut refers to the depth of material removed during each cutting pass.

Its working range is determined by the tool used and the the material type, same as feedrate.

Rougher Cuts should use a higher depth of cut

Finishing cuts should use a lower depth of cut

Tool Material & Characteristics.

To pick a correct tool, it is important to know the material its made of.

To pick a tool with the suitable material, we consider the

Hot Hardness (How well it retains its hardness under high temperatures)

Toughness

Abrasion Resistance

Price/Economical

Acceptable Mechanical Properties

As such, some common tool materials, listed from softest to hardest are

High Carbon Steel/Carbon Tool Steel

Usually not used as its too soft to cut most other materials, which are stronger.

High Speed Steel (HSS)

commonly used in small production batches, and have to be grinded to retain the correct geometry needed.

These tools can also be bought as blanks to be grinded to specific needs

Cast Cobalt Alloys

Cemented Carbide

There are 2 forms of cemented carbide tools:

Throw Away Inserts



These are separate inserts that are fastened onto an empty tool holder, and are the most commonly used.

As there are many different shapes and sizes of inserts, different tool holders are required, with a screw clamp or wedge clamp to fasten the insert.

Most inserts also have a groove/step/obstruction to break up chips.

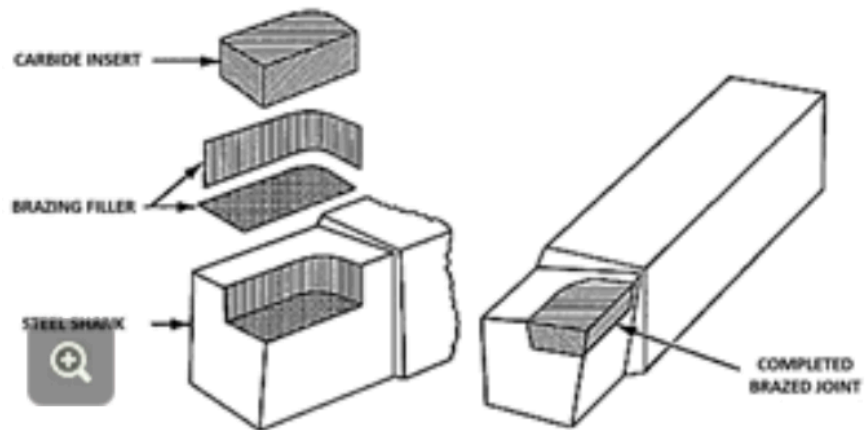
These inserts are also commonly used in mass production, because of how useful they

reduce set up time as the insert can be quickly fastened onto the tool holder,

and are indexable (meaning reusable, by rotating the insert to another cutting edge when one is damaged, for example a triangular insert has 6 cutting edges, 3 on one side, 3 on the other)

Inserts can also be made out of other materials, like ceramic

Carbide Tipped



Carbide tipped tools have the carbide cutting edge is permanently attached to the tool, with the carbide tip being brazed onto a steel shank, giving a hard cutting edge whilst being economical.

Thus, these are more common for small to medium production batches.

Other materials are not used for tips, as these are much stronger than HSS (high speed steel) tools.

Ceramic (as Cemented Oxide)

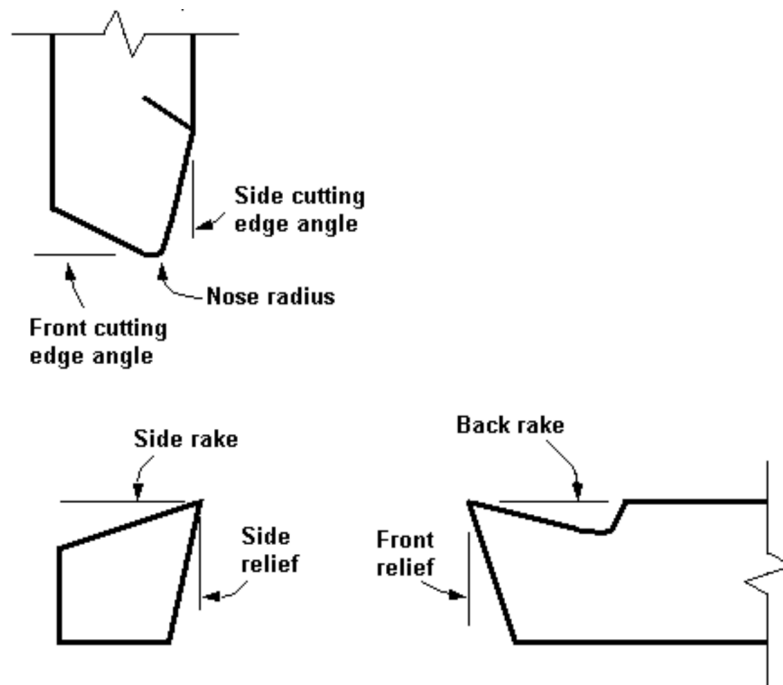
Diamond

usually used as a dust form in grinding wheels

Tool Geometry

Lathe tools have one primary cutting edge (hence called a single pointed tool)

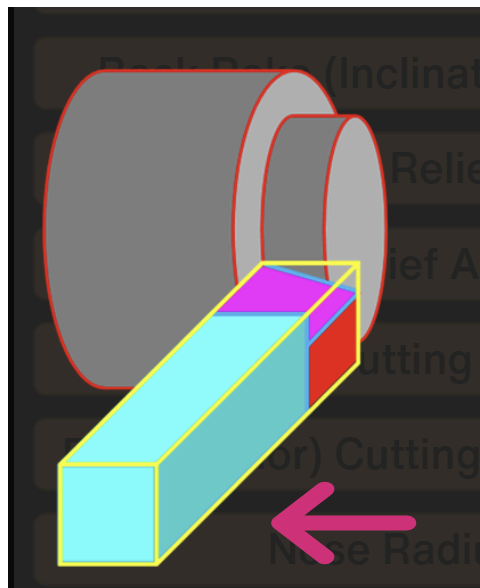
The geometry of these tools have 6 angles, and one radius(rounded corner)



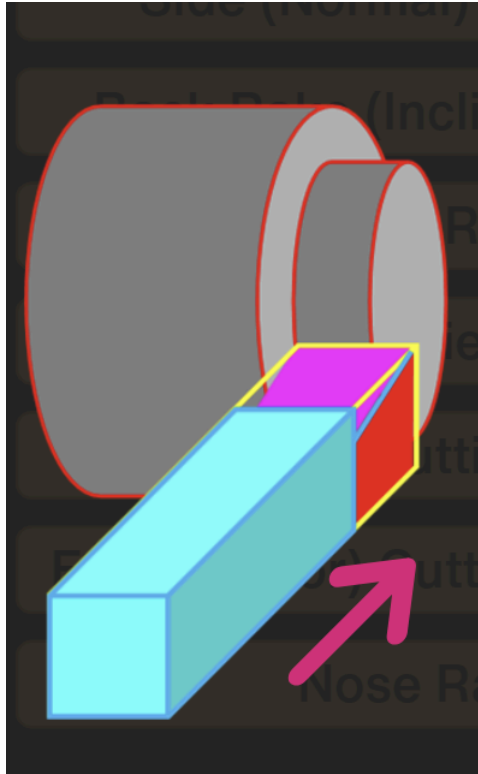
Side/Normal & Back/Inclination Rake Angles

In general, the rake angle allows the tool to cut down the workpiece.

the side rake angle forms a cutting edge for turning operations as well as directing chip flow away from the cut, as shown,



while the back rake angle forms a cutting edge for facing operations, and also directs chip flow away from cut, as shown,

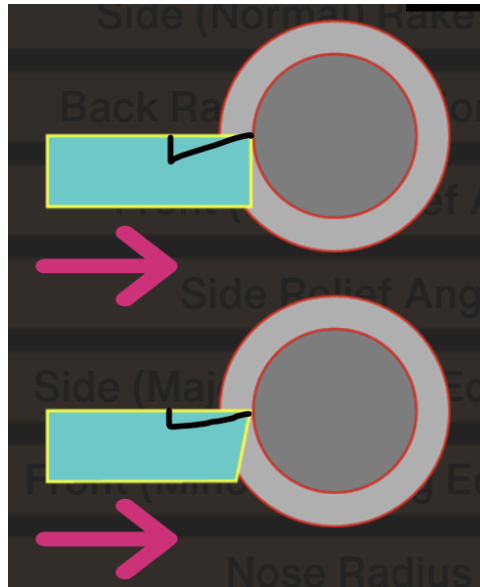


Front/End and Side Relief Angles

In general, Relief Angles allow the tool to be fed into the work while preventing the tool end from rubbing into the FRESHLY cut workpiece surface

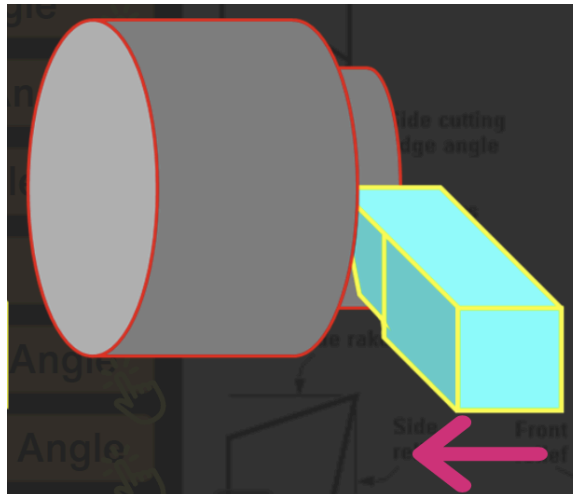
The front relief angle helps to prevent the tool from rubbing against the freshly cut workpiece (cut by the back rake during facing), as shown.

Notice how the front relief angle and the back rake are connected along the same edge.



The Side Relief angle helps to prevent the tool from rubbing against the freshly cut workpiece (cut by the side rake during turning), as shown.

Similarly, the side relief angle and the side rake angle are connected along the same edge.

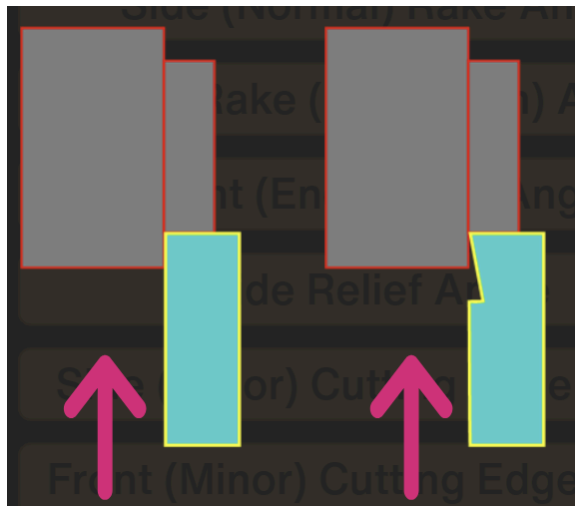


Side/Major and Front/Minor Cutting Edge Angle

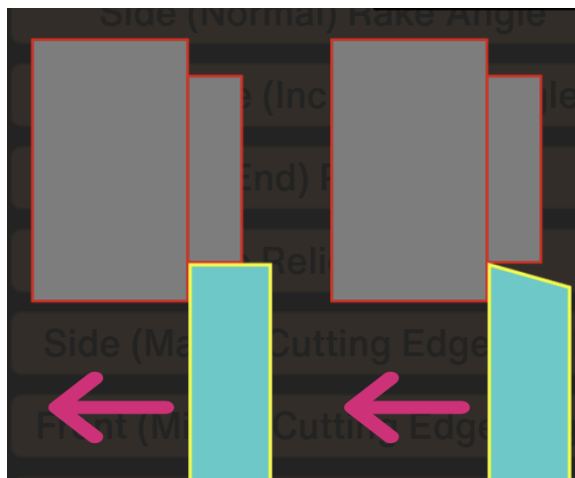
In general, the cutting edge angle defines if the tool is for orthogonal or oblique cutting, as well as providing clearance for the tool to enter the workpiece.

The side cutting edge angle prevents the tool from rubbing against the workpiece surface during facing (that is not freshly cut, i.e. was not in contact with the rake angle edge)

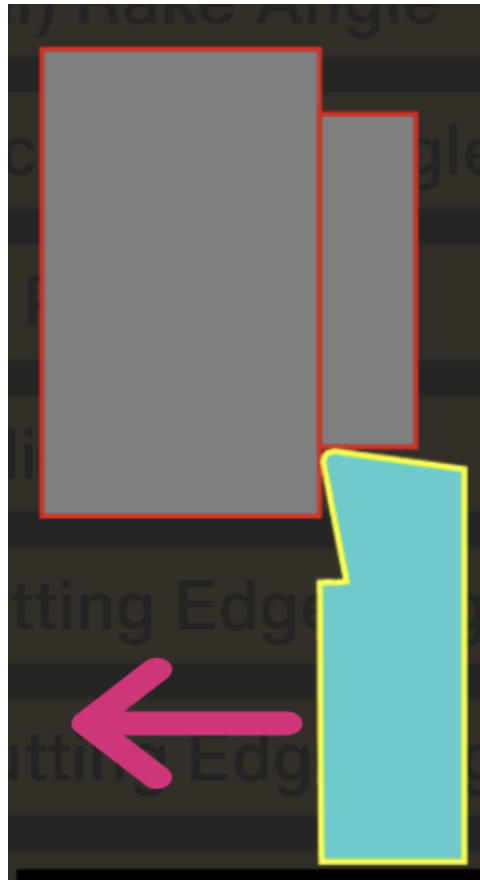
the angle also determines if the cut is oblique or orthogonal??????????????



The front cutting angle prevents the tool from rubbing against the work during turning, as so:



Nose Radius



The nose radius is the rounded corner between the two front cutting angles.

without the nose radius, the sharp tip can break easily or scratch the work, causing poor surface finish.

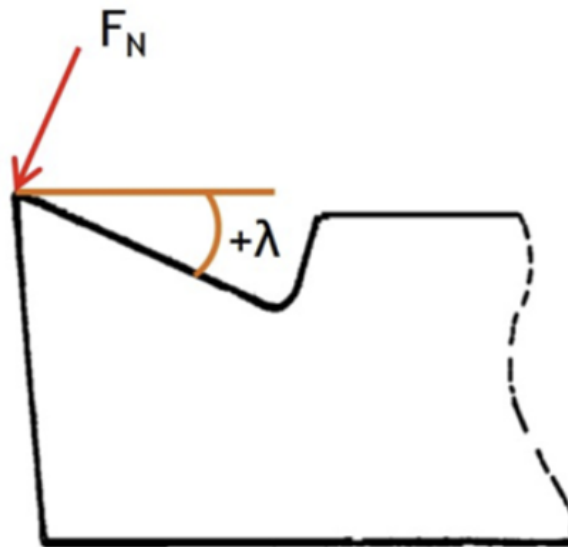
Rounding the tip prevents scratching and strengthens the tip, thus causing a stronger tip and better surface finish.

These angles are combined together to form a cutting tool with surfaces at compound angles.

Positive/Negative Rake Angles

Positive Rake Angle

The rake angle can be positive, meaning the surface is tilted downwards, like so:



For a positive back rake angle, the normal force of the workpiece on the tool acts on the unsupported tool tip, causing the tip to break easily

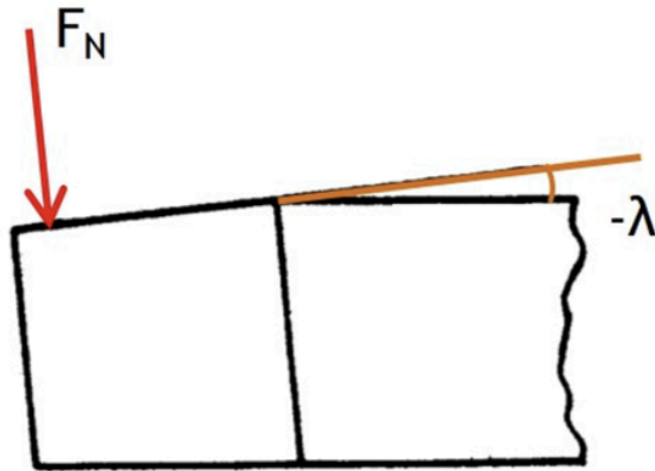
As the rake angle is increased, the tool tip becomes more unsupported, making it weaker.

Because of this, less cutting force and power needs to be used.

However, as materials flow is more fluid, it produces a good surface finish.

Negative Rake Angle

Negative rake angles have the surface tilted upwards, like so:



Because of this, the normal force of the workpiece acts on a supported tool tip, making the tool break less easily.

As the (absolute value) of the angle is decreased, it becomes stronger (as more material is under the tool tip to support the normal force)

This allows higher cutting force and speed, however it does give poorer surface finish (thus making it applicable for tool steel and cast iron, where poor surface finish is not needed)

Cutting Fluids

Cutting fluids, or coolant are usually made from mineral oil.

One such fluid is neat oil, made of 100% mineral oil, useful for slower but higher lubrication operations, like tapping.

Other fluids like emulsions are used, where oil is mixed with water (in different ratios, e.g. 1:20) for faster operations or operations that generate heat, as water has a high heat capacity.

cutting fluids are used for

- cooling and lubrication
- improving surface finish and accuracy
- flushing away chips, preventing chip welding
- prolonging tool life, preventing corrosion
- allowing higher speed and feed rate

Machine Design and Condition

Depending on the type of lathe (e.g tabletop lathe, industrial lathe) have different powers and different purposes.

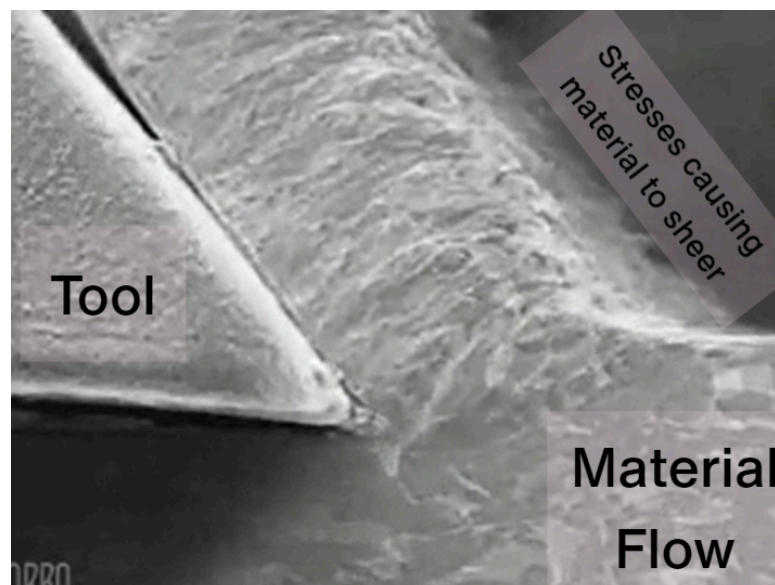
appropriate use and repair frequency affects the machine condition.

Types of Chips

A common way to assess if the cutting method is suitable is by observing the types of chips produced, as they are affected by all the cutting factors (except for machine design)

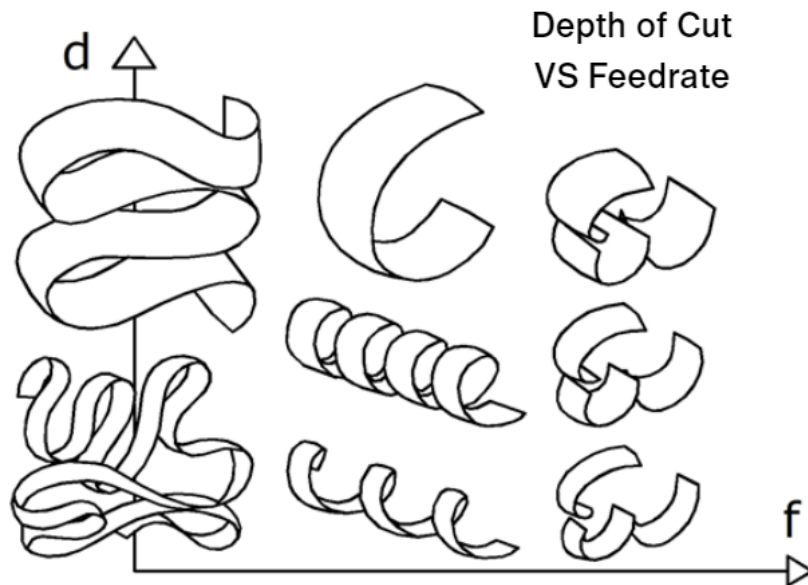
Chip Formation

Chips are formed during the compressive stresses due to cutting.



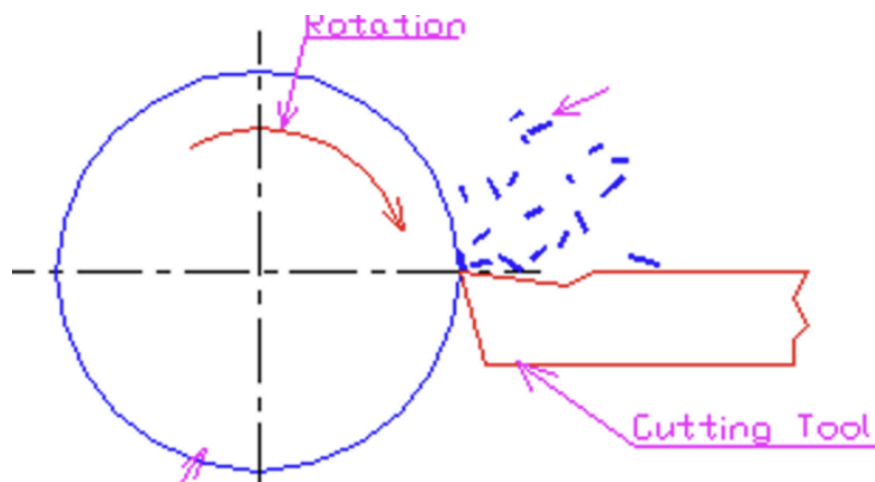
The compressive stresses cause metal movement, which increase until plastic flow (continuous chips) or fracture (discontinuous chips) occur.

This diagram shows how the depth of cut, d and the feed rate, f affect the type of chip produced



Discontinuous Chips

Discontinuous chips are formed before the chip is fully formed, causing them to be segmented and disjointed.

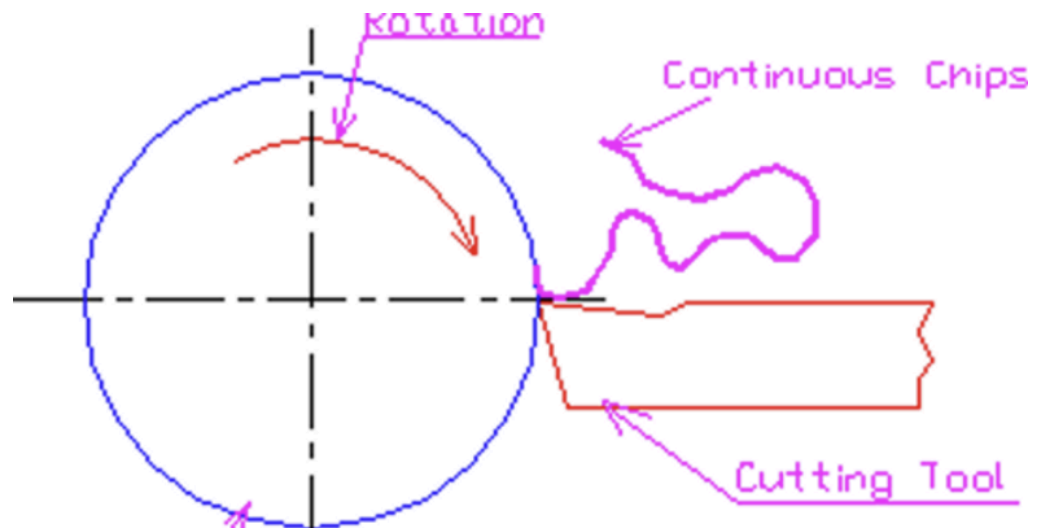


This is mainly due to:

- Brittle work material (e.g. cast iron, leaded brass)
- Small Rake Angle
- Low Cutting Speed
- Large Depth of Cut
- Large Feedrate
- Lack of cutting lubricant

Continuous Chips

Continuous chips form and flow in a continuous manner, forming long and coiled chips.

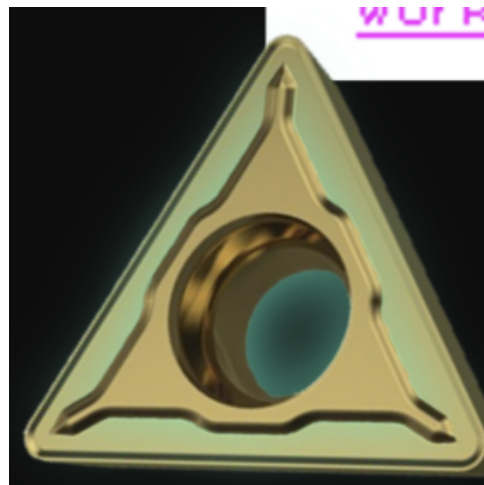


these are due to:

- ductile work material (mild steel, Aluminium)
- high rake angle
- higher cutting speed
- lower depth of cut
- low feedrate
- good use of cutting fluid
- tool has less pressure welding tendency

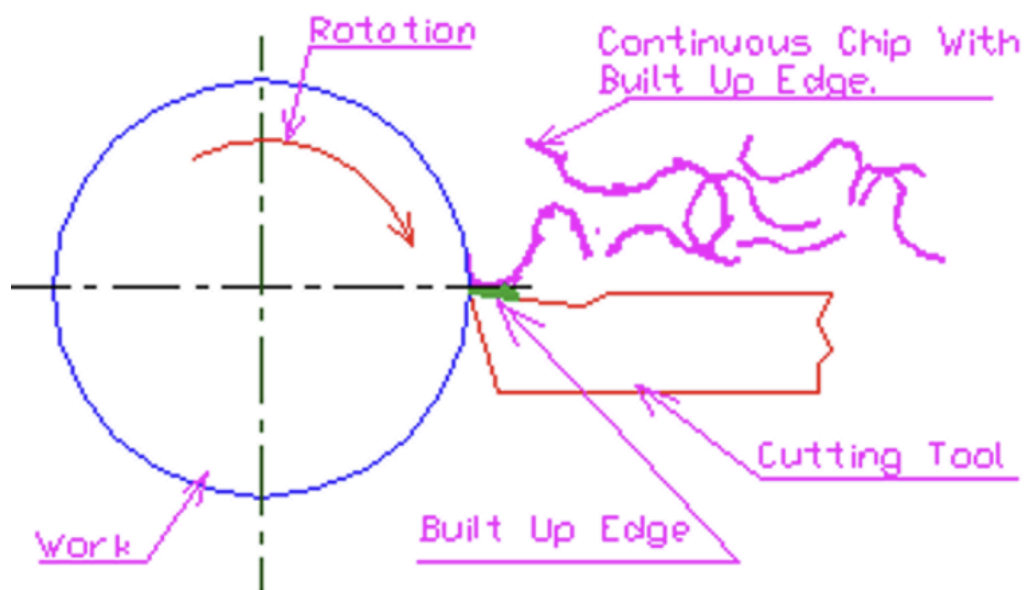
If the chips get too long, it may pose a danger where the chips could get caught in the rotating workpiece (called nesting) or tool.

Thus, inserts with chip breakers are used, to reduce their length.



Continuous Chips with Built Up Edge (BUE)

These chips are formed similarly to continuous chips, however the chip particles are pressure welded to the tools cutting edge.



This is due to:

- ductile work material (e.g. copper)
- low rake angle
- low cutting speed
- large depth of cut
- high feedrate

low/incorrect cutting fluid used (generating friction, thus causing pressure welding)

Feeds & Speeds

Cutting Equations

Recall Cutting Speed and Spindle Speed from Cutting Factors: Cutting Conditions

Spindle Speed, N , is how fast the motor rotates the workpiece, hence is in units rpm

Cutting Speed V , is how fast the workpiece surface "advances", or how fast a point on the surface of the work travels, hence is in units m/min

The two can be connected by the equation:

$$V = \frac{\pi D N}{1000},$$
$$N = \frac{1000 V}{\pi D}$$

where D , the diameter, is in mm.

▼ Derivation

$$\begin{aligned}s[m] &= r[m]\theta[rad] \\ \frac{1}{t[min]}s[m] &= \frac{1}{t[min]}r[m]\theta[rad] \\ V[m/min] &= r[m]\omega[rad/min] \\ V[m/min] &= r[m]2\pi N[rpm] \\ V[m/min] &= 2r[m]\pi N[rpm] \\ V[m/min] &= d[m]\pi N[rpm] \\ V[m/min] &= \frac{D[mm]}{1000}\pi N[rpm] \\ V[m/min] &= \frac{\pi D[mm]N[rpm]}{1000}\end{aligned}$$

Feed rate is the rate the work advances along/into the workpiece.

It can be expressed as how much it moves, in millimeters, for every minute, or for every revolution of the workpiece.

In other words, Feed rate can be expressed in mm/rev $[F_r]$, or mm/min $[F_m]$

The two can be converted like so:

$$F_m = N \times F_r$$
$$\left[\frac{mm}{min} = \frac{rev}{min} \times \frac{mm}{rev} \right]$$

Using the feedrate, the cut time can be calculated.

If the tool needs to advance l mm to make a full cut (e.g. l = length of workpiece for turning, or l = radius of workpiece for facing), the equation for cut time is

$$\text{Cut time} = \frac{l}{F_m}$$

where the cut time is in minutes.

Tool Holding Devices

Recall that tools can be held at either the toolpost on the carriage (e.g. turning tools), or at the tailstock (e.g. drill bits).

Tool posts

Front Tool Post

The front tool post can only hold one tool at a time, making it useful for low batch production.

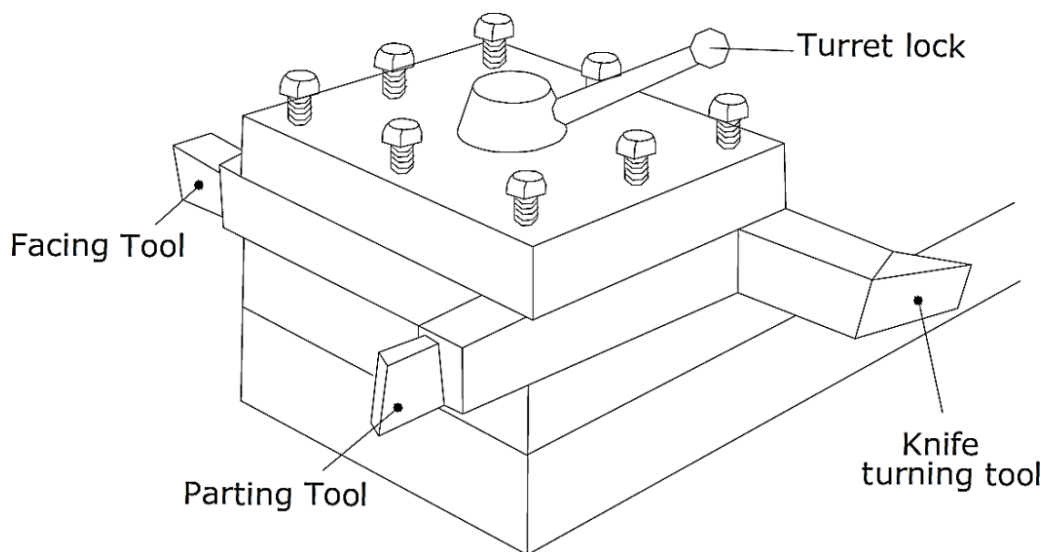
However, it needs packing to achieve center height (i.e. to align tool tip with center), by placing shims (thin metal strips) below the tool to raise it up.



4 Way Toolpost

Recall that 4 way refers to the toolpost being able to hold 4 tools at once.

The tool to be used can be picked by rotating and locking the post into position via an indexing device (labelled in this diagram as a turret lock)



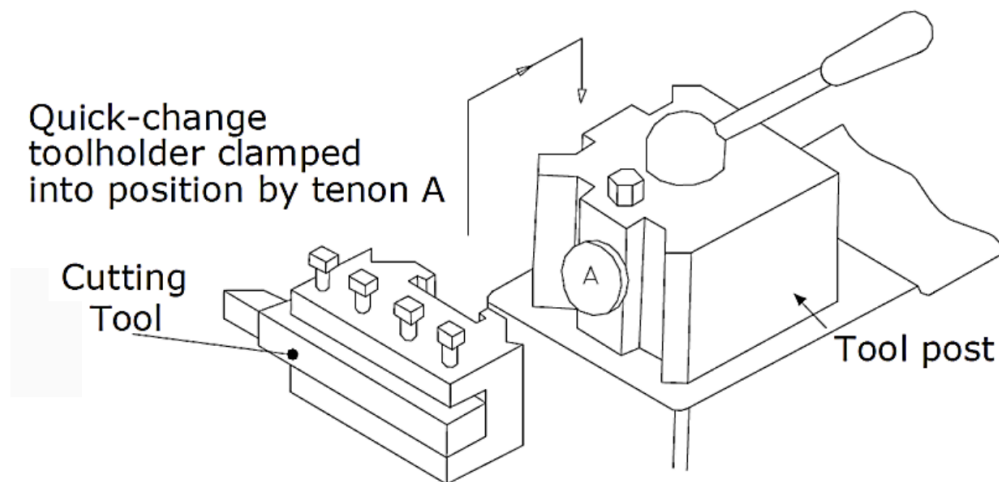
This makes it useful for medium-batch production.

However, like front tool posts, packing is required to achieve center height.

Quick Change Toolpost

Holds 1 or 2 tools, with the tools mounted on holders that can be easily swapped in and out.

The tool center height is also calibrated via an adjustment screw, thus retaining its center height after use, without needing to be recalibrated for next use.



Because of its ease of swapping, and its lack of recalibration, this is useful for high batch production.

However, as each tool requires its own holder, it is the most costly tool holder.

Tool Center Height

To use the tools on the toolpost, the tooltip needs to be inline with the work center.

If the tool is below the center, rake angle decreases and clearance angle increases, becoming inaccurate,

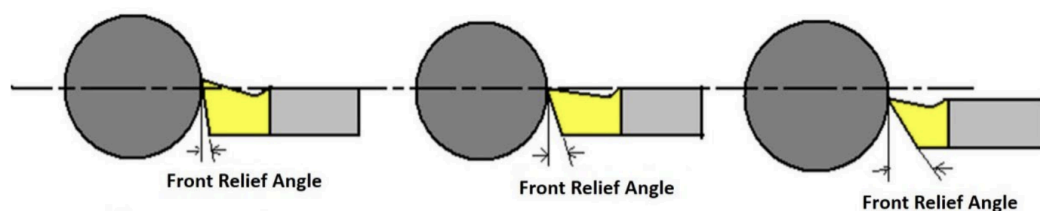
It also causes the tool to dig into the work, causing chatter, and poor surface finish.

If the tool is above the center, rake angle increases and clearance angle decreases, again becoming inaccurate.

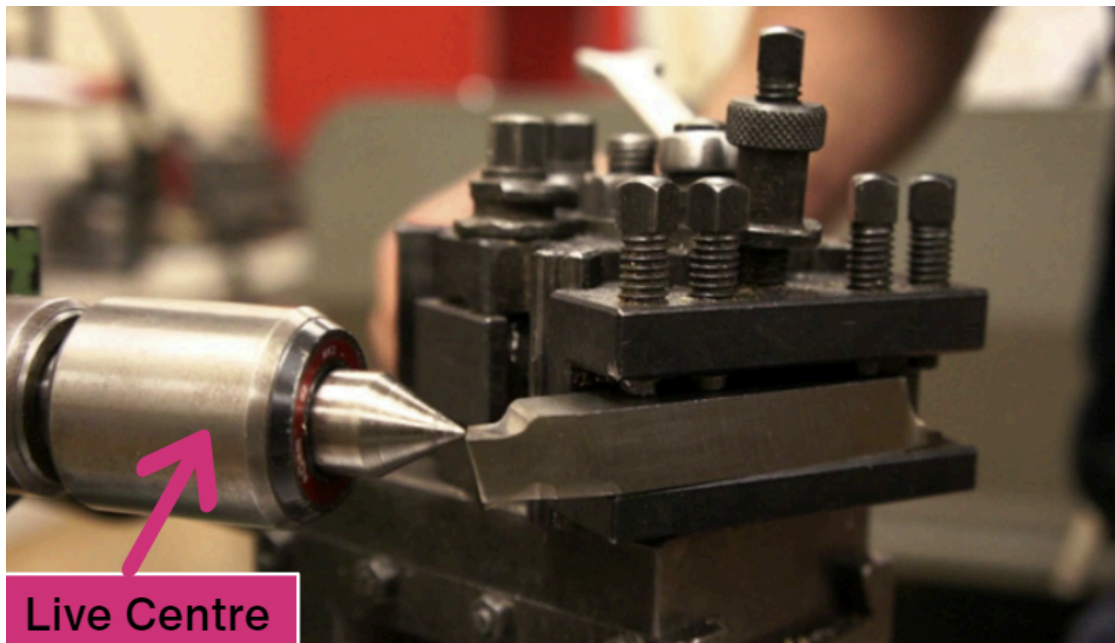
The tool will also rub against the workpiece instead of cutting it..

In general, the cuts will be dimensionally inaccurate when the tool tip and the center are not inline

**rake angle is angle between tool and workpiece, hence it will change when not in center.



Thus, to align the tool tip to the center, the tailstock is mounted with a center as a reference



Tailstock

As said, the tailstock can mount drill bits via a drill chuck.

The drill chuck is placed into the tailstock, and the drill chuck key allows the drill bit to be tightened securely to the drill chuck, for the tailstock to be fed into the work.

The tailstock can also have reaming tools mounted directly to it.

If the tailstock is too big, sleeves can be used as adapter to fit on reaming tools or drill chucks.



Figure 8-8 Sleeve and Reaming Bit

Work Holding & Supporting Devices

A work holding device locates (term for positioning) and rigidly holds the workpiece.

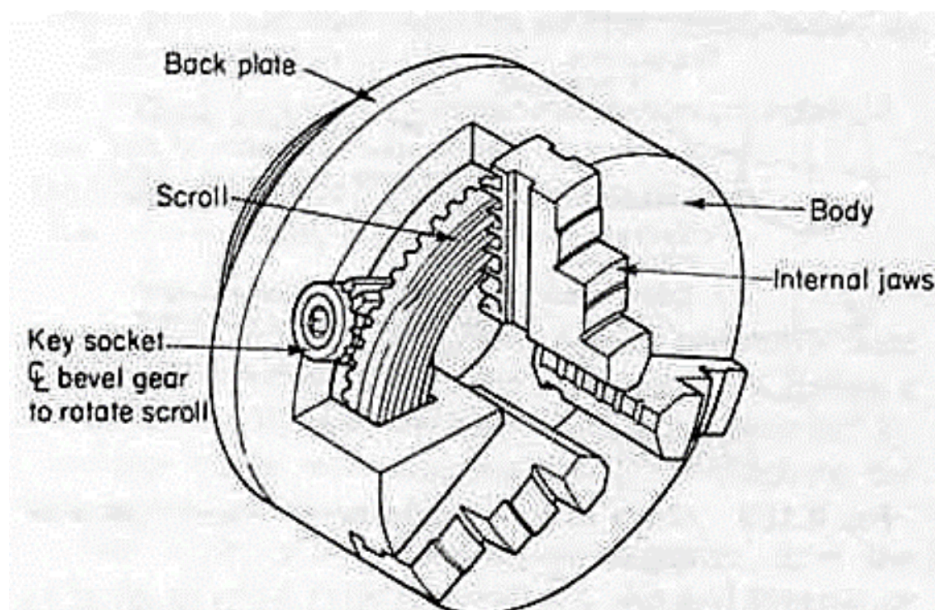
When doing machining work, the work should be done in one set up and should not be removed from the work holding device between operations for measurements/checking/etc, as concentricity will be lost (aka become inaccurate) and needs to be recalibrated again with Dial Gauges.

3 Jaw Chuck

The 3-Jaw Chuck self centers round, triangular, and hexagonal work pieces.

When its key socket is turned, it rotates the scroll, which moves the jaws along the scroll threads, clamping or loosening the work.

As there are only 3 jaws, the gripping force is low as it's split between only 3 points.



The jaws are also removable and reversible, allowing the chuck to hold larger diameter workpieces. However, the jaws are not interchangeable within the slots.

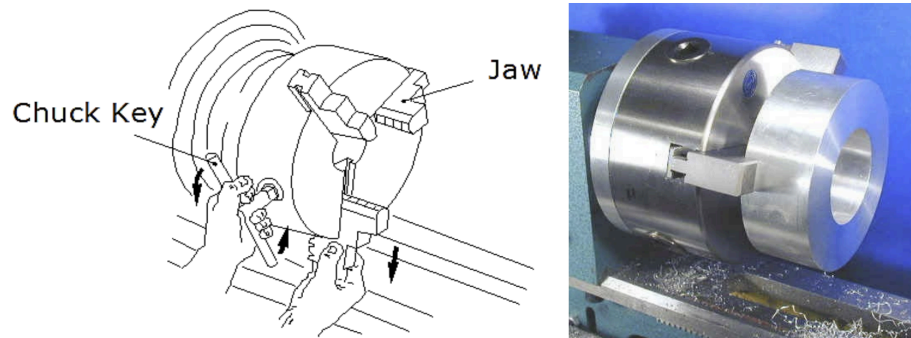


Figure 9-4 Removing Jaws and Large Workpiece Clamping

4 Jaw Chuck

The 4 Jaw Chuck is used for both regular, irregular and off-center workpieces, as each jaw can be adjusted independently with its own key socket.

The independent jaws also make the tightening force stronger, as the tightening force is not split between all 4 jaws and goes to only 1 jaw. The independent jaws also allow higher accuracy, as each jaw can be finely adjusted. However, this makes set up time consuming, and requires dial indicators to ensure the center is in its correct place.

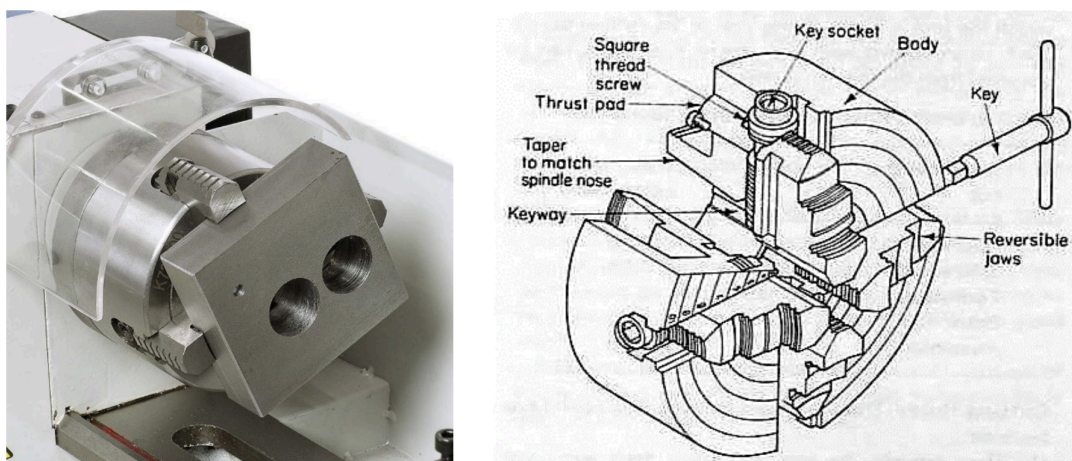


Figure 9-5 4 Jaw Chuck Clamping an Irregular Shape and Its Cut Away

Mandrels

Mandrels hold hollow cylindrical workpieces, like bushes, pulleys, etc., to machine the outside diameter of the work.

A mandrel is essentially a shaft with center holes at each end, and are mounted between centers.

Mandrels are usually hardened and have good surface finish to prevent damage to the work.



There are 4 types of mandrels used, which come in different lengths and diameters

Plain Solid Mandrel

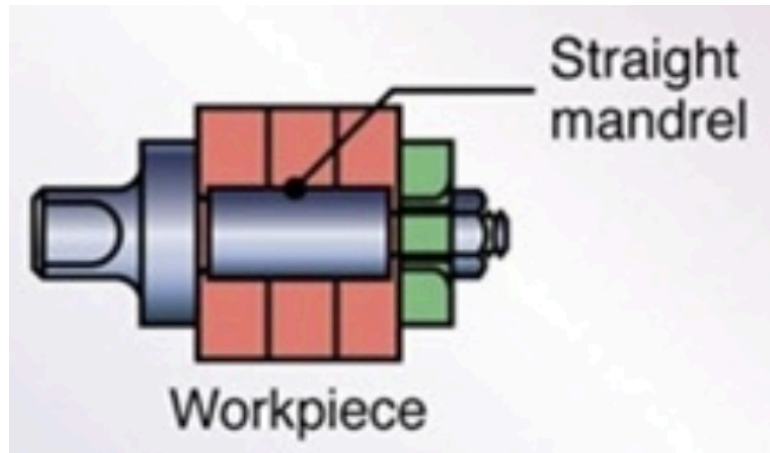
essentially a shaft with one diameter across

Suitable for only one size of hole



Gang Mandrel

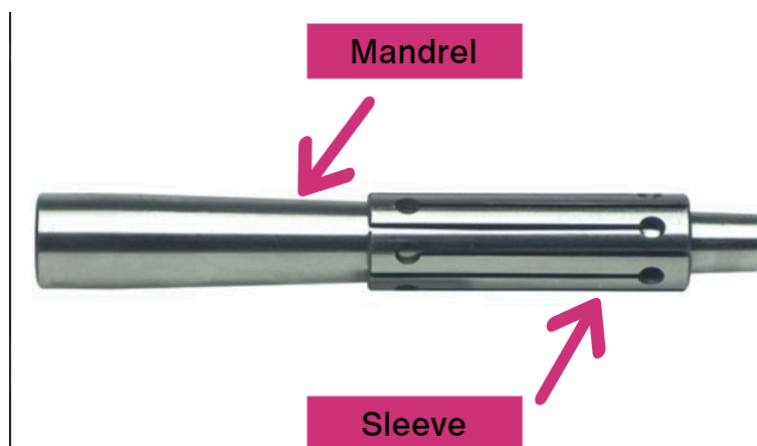
Gang mandrels are useful for holding multiple workpieces of same internal diameter together, held like so



Expansion Mandrel

Has a sleeve whose diameter can expand as its pushed into the tapered mandrel, allowing it to hold onto the workpiece, and can be removed easily

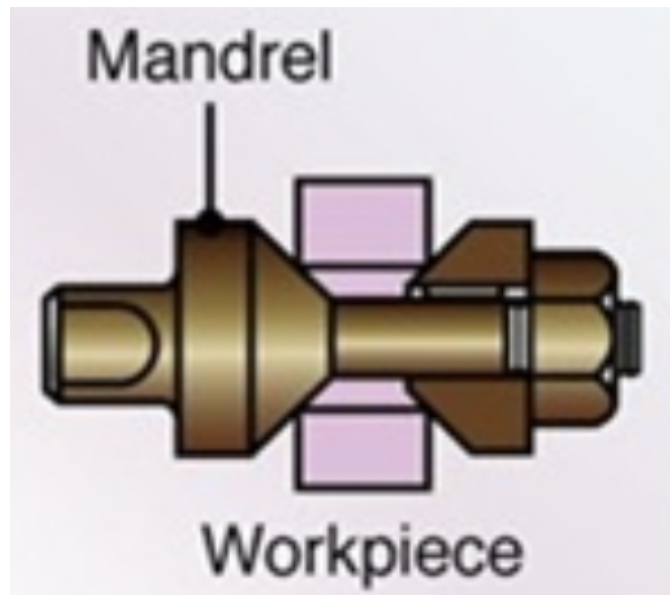
suitable for workpieces with slight variations in internal diameter.



Cone Mandrel

Has a fixed cone and movable cone to hold onto the workpiece like so

suitable for workpieces of different hole diameters, e.g. countersunk holes



For workpieces that are long/thin, they need to be supported to prevent them from flexing under their weight, and to machine them accurately.

A shaft is considered slender if its length/diameter ratio is ≥ 6 , while a shaft is considered elongated if its length/diameter ratio is ≥ 25 .

Thus, different devices are used to support the workpiece

Plain/Dead Centres

Plain centres support the other free end of the workpiece, by being installed in the tailstock.



These are used for general purpose/heavy load/low speed turning.

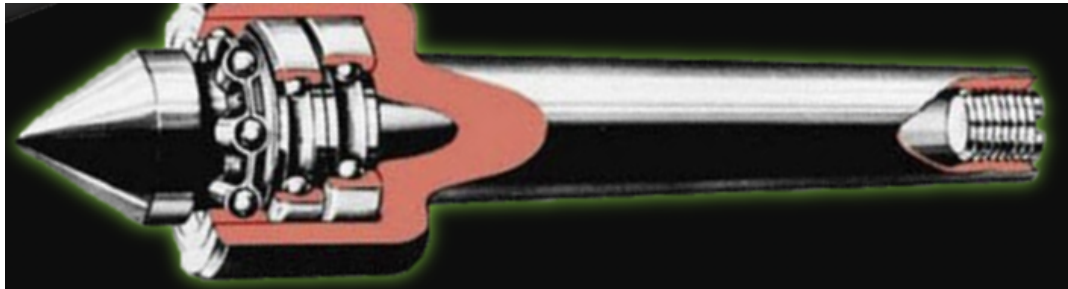
If needed, grease can be applied at the centre's tip to reduce friction.

Like reamers, sleeves may be needed to mount the centre onto the tailstock.

Live/Revolving Centres

Similarly to plain centres, they support the workpiece's free end via the tailstock, however with its ball bearings, its useful for high speed precision turning, as the bearings are able to counter the pressure and heat from heavy cutting

Again, sleeves may be required.



Travelling Steady

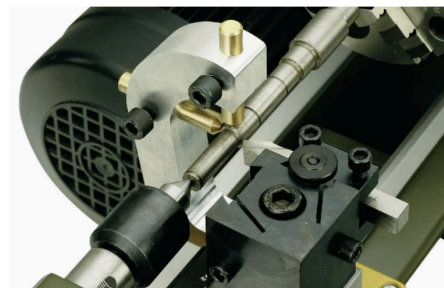
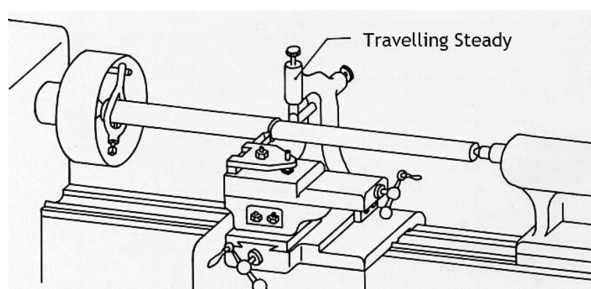
A steady supports the middle of the workpiece/the length of the workpiece.

A travelling steady supports the workpiece as it moves along with the tool, by being bolted to the saddle.

Steadies are used in conjunction with the centre and the chuck supporting the piece at the ends.

The steady itself consists of two point supports which are opposite the tool to counter the cutting forces, and may have the supports greased to reduce friction, or have rollers on the supports to reduce friction too.

These are useful when machining the whole length of the work.



Fixed Steady

Like the travelling steady, it supports the workpiece along its length, however it is bolted onto the bed, and does not move with the tool.

It is used in conjunction with the chuck holding the workpiece in place.

The steady has 3 supports at 120 degrees to each other, with grease or rollers at the supports to reduce friction.

The steady can also be opened up for easy installation of the workpiece.

These steadies are useful for machining an end on a long workpiece, like for facing or boring.

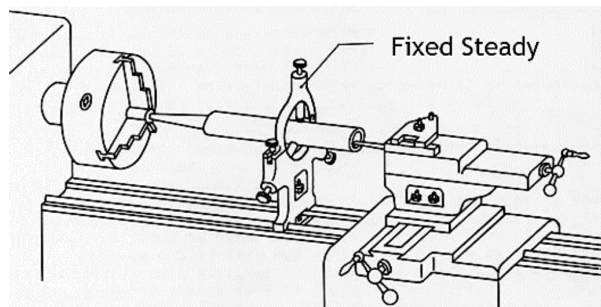


Figure 9-11 Fixed Steady